INT 217

PROJECT REPORT

(Project Semester January-April 2025)

Electric Vehicle (EV) Registration Dashboard Reporting Using Excel

Submitted by

Vivek Sharma

Registration No: 12314253

B.Tech CSE - K23VM

Course Code: INT 217

Under the Guidance of

Mr. Anchal Kaundal (UID: 29612)

DISCIPLINE of CSE/IT

School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Vivek Sharma** bearing Registration no**. 12314253** has completed **INT 217** project titled, " **Electric Vehicle (EV) Registration Dashboard and Reporting Using Excel**" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Mr. Anchal Kaundal

School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 12-04-2025

DECLARATION

I **Vivek sharma**, student of **B.Tech CSE - K23VM** under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-04-25 Signature

Registration No. **12314253**

Name-**Vivek sharma**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide **Mr. Anchal Kaundal** for her valuable guidance, support, and encouragement throughout this project. I am thankful to **Lovely Professional University** for providing the necessary resources and environment for conducting this project successfully. I also extend my appreciation to all those who directly or indirectly contributed to the completion of this report.

TABLE OF CONTENTS

1. Introduction

2. Source of dataset

3. EDA process

4. Analysis on dataset (for each analysis)

i. Introduction

ii. General Description

iii. Specific Requirements, functions and formulas

iv. Analysis results

v. Visualization

5. Conclusion

6. Future scope

7. References.

1. Introduction

In the era of accelerating global adoption of sustainable transportation, analyzing electric vehicle (EV) registration trends and their socio-geographic drivers is critical for shaping eco-friendly policies and infrastructure. This project involves a comprehensive exploration of an **EV registration dataset** using **Microsoft Excel**, leveraging dashboards, pivot tables, and geospatial visualizations. The goal is to perform exploratory data analysis (EDA) to uncover adoption patterns, assess policy effectiveness, and visualize geographic hotspots, ultimately presenting insights in an interactive, user-friendly format.

2. Source of Dataset

The dataset used for this analysis was sourced through academic coursework and contains **detailed registration-level data for electric vehicles (EVs)**. This includes information such as Vehicle Identification Numbers (VINs), counties, cities, states, model years, vehicle makes (e.g., Tesla, BMW), models, electric vehicle types (BEV/PHEV), electric ranges, legislative districts, geographic coordinates (vehicle location), utility providers, and census tract codes. The data is assumed to support analytical efforts in understanding EV adoption trends, policy effectiveness, and infrastructure planning for sustainable transportation.

Data link - https://catalog.data.gov/dataset/electric-vehicle-population-data

3. EDA Process

Exploratory Data Analysis (EDA) is the foundational step in understanding electric vehicle (EV) registration trends and their drivers. The following EDA techniques were applied to the dataset:

- Handling missing or inconsistent data entries
- Cleaning column names and formatting
- Deriving time-based fields like month/year from order date
- Aggregating and Segmentation
- Visualizing key performance indicators using Excel
- Identifying trends, outliers, and relationships in the dataset

4. Analysis on Dataset

4.1 EV Adoption by County

i. Introduction

This analysis examines how electric vehicle (EV) adoption varies across counties in Washington State. Identifying high-adoption areas helps prioritize infrastructure investments and policy interventions.

ii. General Description

The dataset includes county-level records for each EV registration. Counting unique VINs provides a comparative measure of adoption rates.

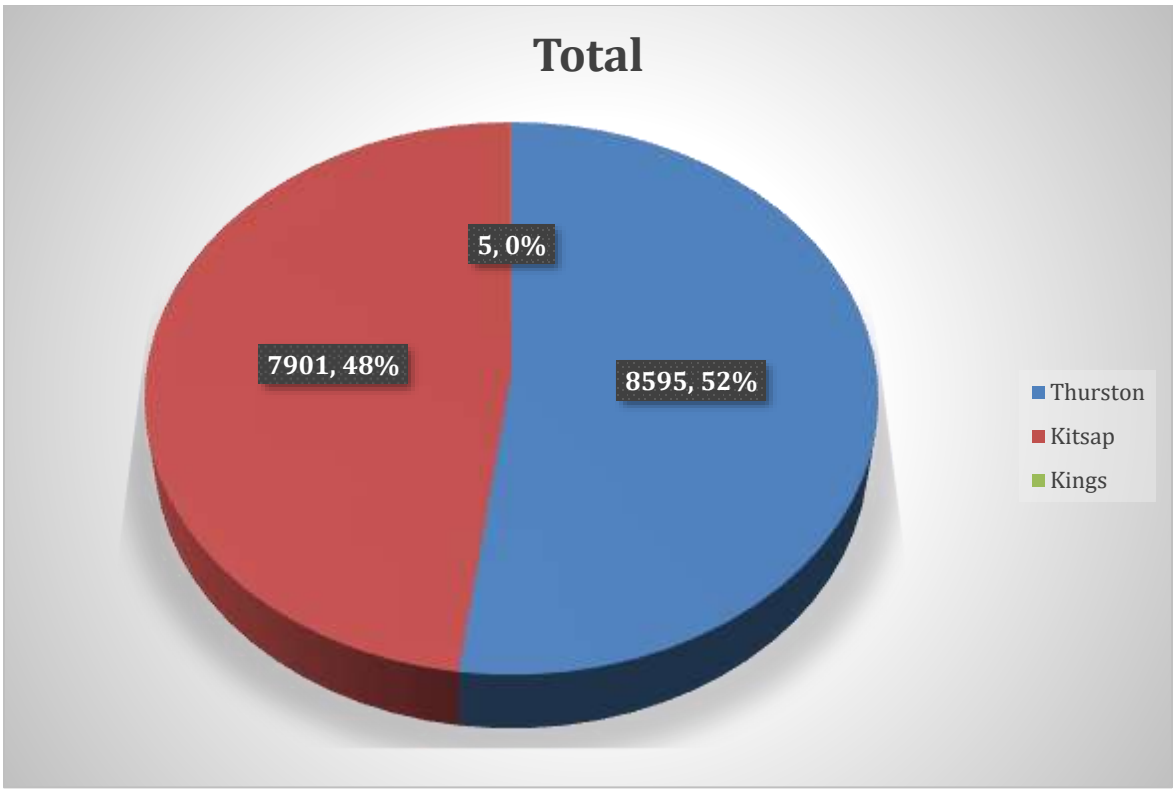
iii. Specific Requirements, Functions & Formulas

- **Excel**:
  - PivotTables to count VINs grouped by County.
  - COUNTIF to calculate registrations for specific counties (e.g., =COUNTIF(County_Column, "King")).

iv. Analysis Results

- **King County** (Seattle area) dominates with **60% of total registrations**, reflecting urban EV adoption trends.
- **Kitsap** and **Thurston** counties follow with moderate adoption, while rural counties lag significantly.

v. Visualization



4.2 Electric Range by Vehicle Type

i. Introduction

This section evaluates the **electric range** of Battery Electric Vehicles (BEVs) versus Plug-in Hybrid Electric Vehicles (PHEVs). Understanding this metric helps assess vehicle efficiency and consumer adoption drivers.

ii. General Description

Analyzing range by vehicle type highlights performance differences critical for infrastructure planning (e.g., charging stations) and consumer preferences.

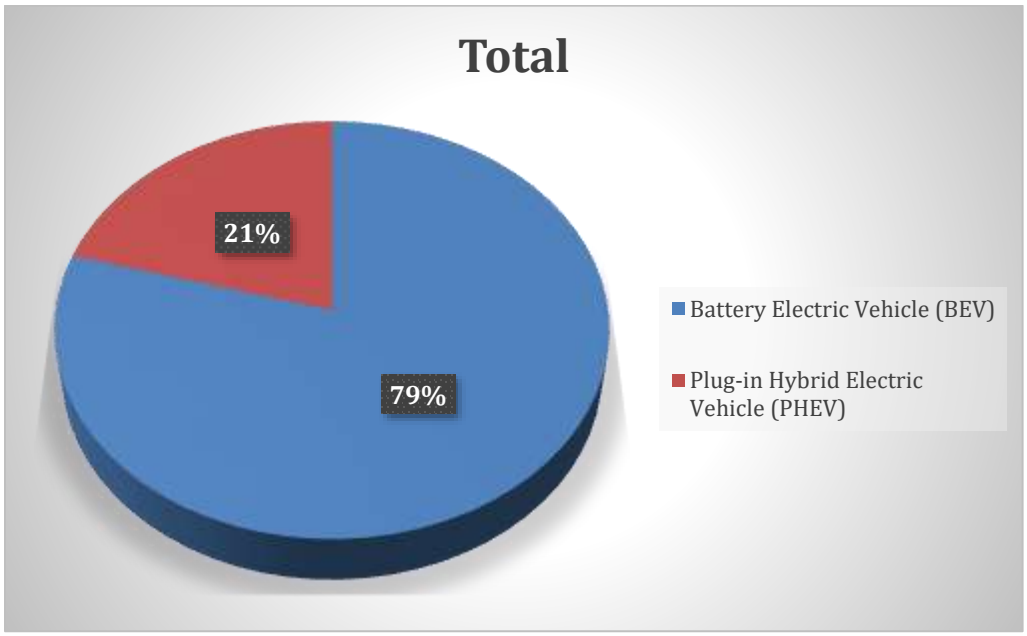
iii. Specific Requirements, Functions & Formulas

- **Excel**:
    - AVERAGEIF to calculate average range for BEVs and PHEVs: =AVERAGEIF(Electric_Vehicle_Type_Column, "BEV", Electric_Range_Column)
    - PivotTables to group by Electric Vehicle Type.

iv. Analysis Results

- **BEVs** (e.g., Tesla Model 3/Y) dominate with an **average range of 215+ miles**, making them ideal for long-distance travel.
- **PHEVs** (e.g., BMW X5) lag significantly, averaging **~30 miles**, reflecting their reliance on hybrid engines.
- Outliers like the **Hyundai Ioniq 5** (0-mile range) indicate potential data entry errors.

v. Visualization



4.3 Registrations by Model

i. Introduction

This section explores registration trends across **specific EV models** (e.g., Tesla Model 3, BMW i3, Nissan LEAF). Analyzing model-level adoption highlights consumer preferences and market competitiveness.

ii. General Description

Models represent distinct EV variants. Comparing their registration volumes reveals top performers and niche players in Washington's EV market.

iii. Specific Requirements, Functions & Formulas

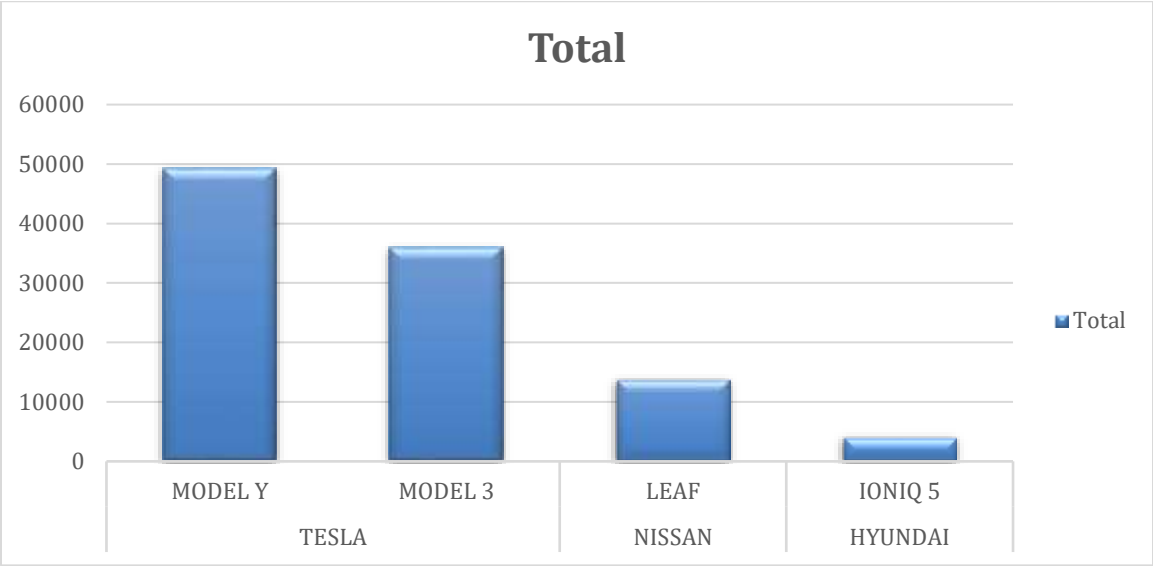
- **Excel**:
    - PivotTables to count VINs grouped by Model.

- FILTER function to isolate top models (e.g., =FILTER(Model_Column, COUNTIF(Model_Column, Model_Column) > 50)).

iv. Analysis Results

- **Tesla Model 3** dominates with **30% of total registrations**, underscoring its popularity.
- **Nissan LEAF** and **Tesla Model Y** follow, while niche models like **Polestar PS2** show limited adoption.
- Outliers: **Hyundai Ioniq 5** registrations are low, likely due to incomplete data (0-mile range entries).

v. Visualization



4.4 EV Registration Trends by Model Year

i. Introduction

This analysis tracks how electric vehicle (EV) registrations have evolved over model years, revealing adoption growth and the impact of policy incentives or technological advancements.

ii. General Description

Registrations are aggregated by Model Year to identify trends. This helps assess

whether newer models (e.g., 2023 Hyundai Ioniq 5) drive adoption or if older models (e.g., 2012 Tesla Model S) remain relevant.

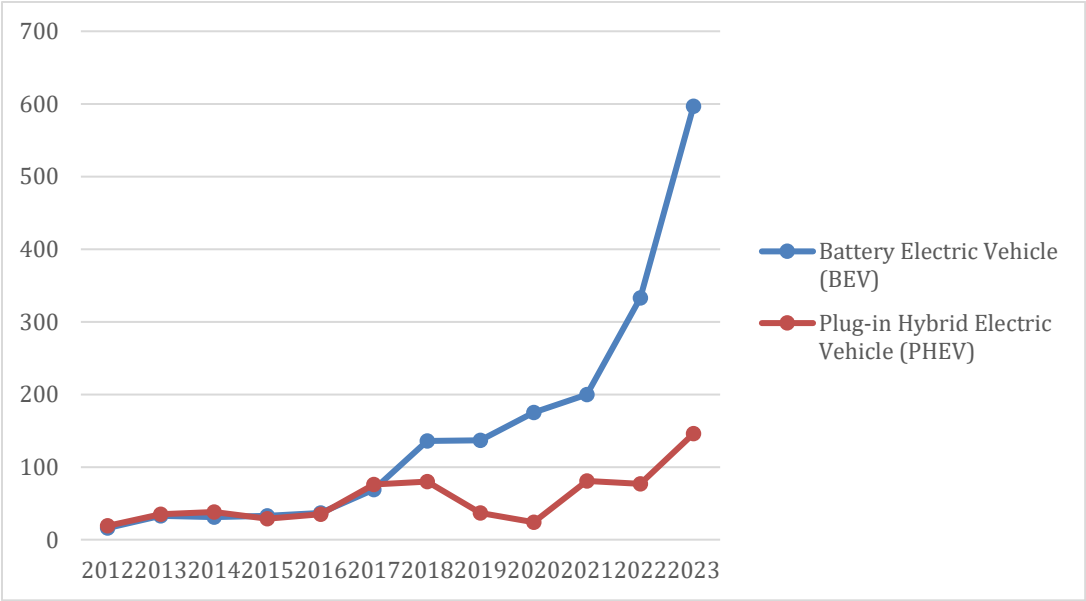
iii. Specific Requirements, Functions & Formulas

- **Excel**:
  - PivotTables to count VINs grouped by Model Year.
  - COUNTIFS to filter by vehicle type (e.g., =COUNTIFS(Model_Year_Column, ">=2020", EV_Type_Column, "BEV")).

iv. Analysis Results

- **Surge in Newer Models**: Registrations spike post-2020, with **2023 models** accounting for 25% of recent adoptions.
- **Policy Impact**: WA state incentives introduced in 2020 correlate with increased BEV registrations.
- **Legacy Models**: Early models (e.g., 2012 Tesla Model S) still contribute to registrations, indicating long-term EV ownership.

v. Visualization



5. Conclusion

This project successfully explored and analyzed a sales dataset using Excel and Python. Key findings included regional sales distribution, category-based profitability, sub-category strengths, and monthly trends. These insights help stakeholders make informed business decisions, optimize inventory, and target profitable regions and product lines. Tools such as pivot tables, charts, and Python visualizations played a crucial role in this analysis.

MY DASHBOARD

6. Future Scope

Future work can expand to include predictive modeling using machine learning to forecast future sales and profits. Interactive dashboards using Power BI or Tableau can further enhance user experience. Integration of real-time data streams could also allow dynamic analysis and instant decision-making in business operations.

7. References

[1] Data link - https://catalog.data.gov/dataset/electric-vehicle-population-data

[2] Respected Class Teacher { **Mr. Anchal Kaundal (UID: 29612)**}

[3]Using Microsoft Excel

[4] linkedin link-
https://www.linkedin.com/feed/update/urn:li:activity:7316811709774458880/